Kubernetes Web Communications Visualization Tool

How I went to this project

- Using AI-enhanced 'VS Code' software (Cursor)
- Believe in integration of AI tools
- Started with personal projects
- Wanted to find a use case for Kubernetes

How I thought of this project

- Lack of "real" mapping of flows in IS: Who is calling who?
- When an error appears… where?
- Logs in our k8s clusters are normalized -> easy to understand in a cluster
  - almost all logs are in json format, it's easy to parse and extract fields

Features (Alpha/Beta)

What it covers:

- Detects webs communications between pods using pod IPs and access logs
  - identify internal and external calls
- Works with one or several k8s contexts
  - A node represents a namespace, or a source if not identified
- Custom rules for log parsing and IP identification
- Multi-threaded: parallelize namespaces analysis in threads to increase performance
- docker image and helm chart to deploy in a k8s cluster

Display and graphical usage:

- **HTTP error detection**: color-coded for 4xx/5xx errors
- Sampling of 5xx errors
- Tooltips on nodes (namespaces/sources) and edges (communications)
- Filtering **by nodes** (highlight for easier visualization)
- Filtering **by error codes**
- Goodie: *Ant mode* 🐜

Why I think it is useful?

- Having a graphical view of flows (Captain Obvious 🦸‍♂️)
- Identifying configuration inconsistencies
  - internal calls with external urls
  - same host called with different http_host
- Quickly locating errors
- But must propose **another view and utility** compared with existing tools like ELK, grafana, loki, etc

Limitations (so far)

- Fetches **the last 'logs lines" for the last minute, every mn, per deployment/namespace**
  - more lines = More CPU
  - more logs collections = More API calls
  - real time... = Fluentbit?
- Custom rules **hardcoded** -> Need YAML or other format
  - inconsistencies in different configurations
- log format must be in json, with extractible fields
  - handle more log formats
  - However, **normalization is powerful**
- **No authentication (yet)**

Technical Enhancements

- extract and enrich with more informations (response_time, last MEP/release, etc)
- **Errors history with databases**
- Make **custom rules easier** with config files (get rid of custom rules within the code and normalization of logs format)
- Add more IP rules to include **externals components**
- Separated and asynchronous systems for logs collections and analyzis
    - increase logs collection and buffering before analyzis
- Reduce API calls to k8s by building a enriched dictionnary
- code refactoring to ease comprehension:
    - python almost ok, but needs cleaning
    - separate js script

Usage enhancements:

- Make filtering more efficient
- Improve or tune physics for more visibility
- add links in tooltips to another tool for details?

Learnings and thinkings working with AI

- **GIT! GIT! (Did I say git?)**
- AI tools can **boost efficiency**
- AI helps with **initial setup**, but struggles with complexity
  - we mustn't be lazy, asking too much can lead to more mess
  - bugs could appears very easily (AI does not remind and understand all the code it has produced before, or it's expensive! :))
- **Think before using AI**: understand, correct, and guide the AI
- Easy to get lost → **Know and organize your code!**

Tools used:

- Cursor = VS Code powered with AI (with Anthropic/GPT4)
- LocalAI (local engine to provide LLMs internally)
  - less performant than "pro online models" like openAI/Anthropic/Gemini/Phi etc but nothing goes outside (models available gpt4, phi4, falcon, qwen2.5 (alibaba), stablediffusion, deepseek vX, etc)
  - ability to test several LLMs
- online AI tools

Resources I want to explore:

- Customizing prompts
- Tools like k8sgpt, with custom filters
- LLM backdoor (not related to this project, just readings - tool to enrich an existing LLM - Training for logs?)
  - using the tool to customize a model => more a ML engineer job?
- https://medium.com/@rehmana.younis1/title-building-an-advanced-log-analyzer-chatbot-with-llms-rag-and-streamlit-8b8a203487c0
- https://mudler.pm/posts/localai-question-answering/
  - Chroma/LangChain/RAG (Retrieval-Augmented Generation) and localAI to analyse documents.
- security working with AI
  - despite the performance, that's why I used an internal server